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CARBON-ION RADIOTHERAPY FOR LOCALLY ADVANCED OR UNFAVORABLY LOCATED CHOROIDAL MELANOMA: A PHASE I/II DOSE-ESCALATION STUDY

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Purpose: To evaluate the applicability of carbon ion beams for the treatment of choroidal melanoma with regard to normal tissue morbidity and local tumor control.

Methods and Materials: Between January 2001 and February 2006, 59 patients with locally advanced or unfavorably located choroidal melanoma were enrolled in a Phase I/II clinical trial of carbon-ion radiotherapy at the National Institute of Radiologic Sciences. The primary endpoint of this study was normal tissue morbidity, and secondary endpoints were local tumor control and patient survival. Of the 59 subjects enrolled, 57 were followed >6 months and analyzed.

Results: Twenty-three patients (40%) developed neovascular glaucoma, and three underwent enucleation for eye pain due to elevated intraocular pressure. Incidence of neovascular glaucoma was dependent on tumor size and site. Five patients had died at analysis, three of distant metastasis and two of concurrent disease. All but one patient, who developed marginal recurrence, were controlled locally. Six patients developed distant metastasis, five in the liver and one in the lung. Three-year overall survival, disease-free survival, and local control rates were 88.2%, 84.8%, and 97.4%, respectively. No apparent dose–response relationship was observed in either tumor control or normal tissue morbidity at the dose range applied.

Conclusion: Carbon-ion radiotherapy can be applied to choroidal melanoma with an acceptable morbidity and sufficient antitumor effect, even with tumors of unfavorable size or site.

Carbon ion radiotherapy, Choroidal melanoma.

INTRODUCTION

Malignant melanoma is a relatively radio-resistant tumor, although high-dose, well-localized radiotherapy with proton beams or treatment with radioactive plaques has shown satisfactory results in patients with choroidal melanoma, especially in small to medium-sized tumors (1–3). The National Institute of Radiologic Sciences (NIRS) was the only facility in Japan, performing proton radiotherapy (PRT) for choroidal melanoma. It opened in February 1986 and treated 92 patients with the NIRS cyclotron through March 2003. The results were satisfactory, comparable to those reported from overseas PRT facilities (1–3).

The Heavy Ion Medical Accelerator in Chiba (HIMAC) was built at the NIRS in 1994, and the clinical studies of carbon ion radiotherapy (C-ion RT) at various tumor sites have been conducted successfully (4, 5). Carbon ion beams have unique physical and biological properties that make well-localized, high linear energy transfer (LET) radiotherapy possible (6). For choroidal melanoma, high-dose radiation with excellent dose conformity is essential to obtain satisfactory results as has been proved by PRT and plaque therapy; therefore, C-ion RT was thought to be suitable for the treatment of this tumor. Moreover, high LET components of carbon ion beams offer a sufficiently effective treatment option that may achieve comparable tumor control probability with a lower dose than that applied in the PRT. A Phase I/II dose-escalation study of C-ion RT for choroidal melanoma was carried out to investigate the usefulness of carbon ion beams for this tumor type in terms of tumor

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